Diskret Matematik Formelsamling

1 Inclusion-Exclusion

1.1 – General

For n=2

$$|A \cup B| = |A| + |B| - |A \cap B|$$

For n=3

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

For $n = 4$

$$\begin{aligned}|A \cup B \cup C \cup D| = &|A| + |B| + |C| + |D| \\ &- |A \cap B| - |A \cap C| - |A \cap D| - |B \cap C| - |B \cap D| - |C \cap D| \\ &+ |A \cap B \cap C| + |A \cap B \cap D| + |A \cap C \cap D| + |B \cap C \cap D| \\ &- |A \cap B \cap C \cap D|\end{aligned}$$

Generally for n sets

If $A_1, A_2, \ldots, A_n$ are n finites sets, then:

$$\begin{aligned}|A_1 \cup A_2 \cup \cdots \cup A_n| = &\sum_{1 \leq i \leq n} |A_i| \\ &- \sum_{1 \leq i \leq j \leq n} |A_i \cap A_j| \\ &+ \sum_{1 \leq i \leq j \leq k \leq n} |A_i \cap A_j \cap A_k| \\ &- \cdots \\ &+ (-1)^{n+1} |A_1 \cap A_2 \cap \cdots \cap A_n|\end{aligned}$$

1.2 – Inclusion-Exclusion of properties

For x elements we have n properties $P_1, P_2, \ldots, P_n$. We say that P_i' is the negation of property P_i.
The number of elements satisfying all properties P_i, P_j, P_k is $N(P_i, P_j, P_k)$

The number of elements that satisfy *none* of the properties is:

$$\begin{aligned} N(P_1', P_2', \ldots, P_n') = N &- \sum_{1 \le i \le n} N(P_i) \\ &+ \sum_{1 \le i < j \le n} N(P_i, P_j) \\ &- \sum_{1 \le i < j < k \le n} N(P_i, P_j, P_k) \\ &+ \cdots \\ &+ (-1)^n N(P_1, P_2, \ldots, P_n) \end{aligned}$$

where N is the total number of elements.

1.3 – Derangements

A derangement is a permutation that leaves no object in its original position.
The amount of derangements of n objects, denoted by $!n$, is given by:

$$D_n = n! \left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots + (-1)^n \frac{1}{n!} \right]$$

10 Graphs

Table 1: Graph Terminology

Type	Edges	Allow Multiple Edges?	Allow Loops?
Simple graph	Undirected	No	No
Multigraph	Undirected	Yes	No
Pseudograph	Undirected	Yes	Yes
Simple directed graph	Directed	No	No
Directed multigraph	Directed	Yes	Yes
Mixed graph	Both	Yes	Yes

Multiple edge mean multiple edge between the same pair of vertices. A loop is an edge that connects a vertex to itself.

10.1 – Basic knowledge

The sum of the degrees of all vertices is 2 times the number of edges. Therefore the sum is always even.
This means the number of vertices with odd degree is always even.
The sum of in-degrees is equal to the sum of out-degrees, and both are equal to the number of edges.

10.2 – Common Graphs

Complete Graphs K_n

Graph with one edge between each pair of vertices.
Vertices: n **Edges:** $\frac{n(n-1)}{2}$ **Degrees of vertices:** $n-1$

Cycles C_n

Graph where the vertices are connected in a closed chain. Only defined for $n \geq 3$.
Vertices: n **Edges:** n **Degrees of vertices:** 2

Wheels W_n

A cycle with an additional center vertex connected to all other vertices. Only defined for $n \geq 4$.
Vertices: n **Edges:** $2(n-1)$ **Degrees of vertices:** Center: $n-1$, Other: 3

n-cubes Q_n

A cube of dimension n. A cube of dimension n can be made by connecting the corresponding vertices of two cubes of dimension $n-1$.
Vertices: 2^n **Edges:** $n2^{n-1}$ **Degrees of vertices:** n

11 Polynomials

An n'th degree polynomial can be written as $P(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0$ where $a_n \neq 0$. The degree of the polynomial is n.
$[x^k]P(x)$ denotes the k'th degree coefficient. So for $P(x) = 3x^2 + 4x + 2$ then $[x^1]P(x) = 4$

Algebra

Addition:

$$(ax^2 + bx + c) + (dx^2 + ex + f) = (a + d)x^2 + (b + e)x + (c + f)$$

Which means that:

$$[x^k](P(x) + Q(x)) = [x^k]P(x) + [x^k]Q(x)$$

Scalar multiplication:

$$c \cdot 4x^2 + 7 = (c \cdot 4)x^2 + (c \cdot 7)$$

Which means that:

$$[x^k](cP(x)) = c[x^k]$$

Multiplication: For $P(x) = ax^3 + bx$ and $Q(x) = cx^3 + dx^2 + ex + f$:

$$P(x)Q(x) = (ac)x^6 + (ad)x^5 + (ae + bc)x^4 + (af + bd)x^3 + (be)x^2 + (bf)x$$

Which means that:

$$[x^k](P(x)Q(x)) = \sum_{k=0}^{n} [x^k]P(x) \cdot [x^{n-k}]Q(x)$$

11.1 – Other info

The division algorithm for polynomials states that for two polynomials $P(x)$ and $D(x)$, there exist unique polynomials $Q(x)$ (the quotient) and $R(x)$ (the remainder) such that:

$$P(x) = D(x)Q(x) + R(x)$$

where deg$(R(x)) < deg(D(x))$ or $R(x) = 0$.

Theorem 6.1